

# Evaluating Record-Level Supervisory Analytics for Security Operations Centres: Controlled Validation, Decision Integrity

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**Abstract.** Supervisors of security operations centres must judge periodically whether monitored organizations handle security work properly, yet their usual instruments are self-assessed maturity models and manual record review. We study whether the records an organization submits can instead be analysed into findings that cite those records, ranked across organizations, decided by a human supervisor, and bound to that decision cryptographically. We evaluate this workflow as implemented in SAT-SA, whose 16 deterministic, rule-based analytical workers form the evaluated subset of 32 registered components, using frozen, hash-verified experiment bundles. On controlled data the mechanisms behave as designed: every declared finding family was emitted, validation matched its declared contract in all 245 perturbed conditions, all 13 tested mutations of the decided record were detected, and all 36 injected interruptions recovered. Simple rules are competitive: fixed-threshold and median-deviation rules tied the fast-closure detector, and on generated populations that favour SAT-SA by construction a closure-speed heuristic nearly matched its ranking; stale and contradictory records were accepted silently. On a public IT incident log, which is not security-operations data, the risk score was not associated with service-level misses (Spearman  $\rho = -0.11$ ), whereas slowest median resolution was ( $\rho = 0.94$ ); a failure analysis attributes this to construct mismatch and detector saturation. The evidence supports mechanism-level claims, not effectiveness in real security operations, for which no human study or expert labels exist.

**Keywords:** Security operations · Supervisory analytics · Incident management · Process conformance · Human-in-the-loop decision support · Post-quantum signatures · Negative results.

## 1 Introduction

Security operations centres (SOCs) triage large alert volumes with limited analyst capacity [47,45], and field studies document recurring process problems: false-positive-dominated alarms, mismatched expectations between analysts and management, and burnout [5,24,43]. When an external body supervises many SOC—for example a national authority for critical infrastructure—it must judge periodically whether each monitored organization actually investigated, escalated and closed its work properly. The instruments available are mostly maturity self-assessments such as SOC-CMM [30] and CTI-SOC2M2 [39], metric catalogues [16,4], and manual examination of records. Self-assessment reports capability, not conduct; manual examination does not scale across many organizations and periods.

Record-based assessment of incident handling is not new. Palma et al. assess the compliance of an incident-management process with a reference model derived from ITIL and ISO/IEC 27035 by aligning its event log with that model and prioritizing the deviations for an auditor [32,35], and support the assessor with visual analytics [33,34]. SOC performance has also been measured empirically by injecting synthetic attacks into an operational SOC [37]. What remains open for the supervisory setting is narrower: an analysis of the heterogeneous records a monitored SOC submits (alerts, cases, investigation steps, escalations, dispositions, assets) that yields findings citing those records, ranks organizations against each other, keeps a human supervisor as the decision authority, and makes the decided state verifiable afterwards—together with an honest account of how such an analysis behaves on controlled and on independent data.

This paper studies that question with an implemented system, SAT-SA (developed for the Smart India Hackathon problem SIH26157), and a frozen evaluation. Its emphasis is the evidence rather than the architecture: every number is taken from a hash-verified experiment bundle, and negative findings are reported with the same prominence as positive ones.

*Contributions.* We separate contribution types and state their strength.

- *Application framing (low-confidence candidate)*. Supervisor-side, record-level analytics over SOC evidence submissions, with findings that cite records, cross-organization peer comparison, entity ranking and a bound human decision. Each ingredient has prior art (Section 2); the combination is offered without a novelty claim.
- *Empirical findings*. Controlled evaluations of detection against declared labels and simple baselines, one-worker ablation, imperfect-evidence robustness, peer-cohort sensitivity, orchestration cost and recovery, and an external negative result on a public IT incident log with a construct-level failure analysis.
- *Engineering*. A tenant-aware submission, analysis and review workflow with a durable queue and a checkpointed human-review interrupt.
- *Security capability (integration)*. Binding of the supervisory decision to the analysed state with standard primitives (SHA3-256, ML-DSA-65, a hash-chained ledger), evaluated with a mutation matrix.
- *Reproducibility practice*. Write-once experiment bundles, an evidence freeze and a data layer that generates every reported number.

We make no claim of algorithmic novelty, superiority over existing systems, or effectiveness in real SOC.

The remainder of the paper reviews related work (Section 2), formalizes the problem and the implemented analysis (Sections 3 and 4), describes the experimental methodology (Section 5), reports controlled results (Section 6) and the external evaluation (Section 7), and discusses implications, threats to validity and reproducibility (Sections 8–10).

## 2 Related Work and Research Gap

*Search method.* We positioned the work with a structured search, not a systematic review. Twenty queries on SOC assessment, alert prioritization, security process mining, peer analytics, supervisory technology and tamper-evident records were run on OpenAlex and Crossref (656 unique records; one screener; 106 abstracts read; 51 included and joined with 26 sources of an earlier focused review), followed by an update search for closest prior art and counterevidence. arXiv throttled the automated queries and records beyond rank 25 per query were not screened, so absence from this review is not evidence of absence. Queries, counts and decisions are in the supplementary material.

*SOC assessment and measurement.* Maturity models score capability through self- or expert assessment [30,39]; metric work defines and weights indicators of analyst or technical performance [3,4,16]. A case study in one academic SOC found that analysts who scored well on key performance indicators did not necessarily further the SOC’s goals [27], a caution for any metric-based supervision. SAIBERSOC measures an operational SOC empirically by injecting synthetic attacks and observing outputs such as detection accuracy and time to investigation, in an experiment with 124 students acting as analysts [37]. These approaches assess capability or test a SOC actively; none analyses the records a SOC submits to an external supervisor.

*Record-based assessment of incident handling (closest prior art).* Palma et al. built a system that assesses an organization’s incident-management process for compliance with a reference process derived from ITIL and ISO/IEC 27035: it aligns the event log with the model, classifies deviations with a taxonomy, prioritizes them with a cost model and supports an auditor; it was validated on a benchmark and demonstrated on a public real incident-management log [32]. A companion benchmark compares such assessment models [35], and two visual-analytics tools support the human assessor [33,34]. The public log used in that line of work is the ServiceNow incident log (141,712 events over 24,918 incidents) released with [7] and published as [6]—the same log we use for our external evaluation, and one on which SLA attainment has also been modelled directly [44]. Automated, record-based assessment of incident handling for an assessor therefore already exists and has been demonstrated on our external data. Table 1 contrasts it with SAT-SA.

*Alert and incident prioritization.* A large literature ranks alerts or incidents for analysts [21,45]: resource-aware selection of alerts to investigate [41], provenance-based triage [20], guided response learned from customer triage labels [17], learning to defer to analysts [22], and, recently, LLM agents embedded in SOC triage [8]. DomainPrio analyses the population of a SOC’s investigations rather than single alerts [12]. SAT-SA differs in unit and user—it ranks monitored organizations for a supervisor from findings about how work was handled—and is not compared with these systems.

Table 1. Closest prior work compared with SAT-SA, as described in the cited sources (“–” not addressed; “n/d” not determinable from the source).

| Work                          | Evidence                                      | Analysis                                       | Across orga-<br>nizations | Human role;<br>decision<br>binding                        |
|-------------------------------|-----------------------------------------------|------------------------------------------------|---------------------------|-----------------------------------------------------------|
| Maturity<br>models<br>[30,39] | self or expert<br>assessment                  | maturity<br>scoring                            | comparable<br>scores      | assessor; –                                               |
| SAIBERSOC<br>[37]             | outputs of an<br>operational<br>SOC           | synthetic<br>attack<br>injection               | SOC config-<br>urations   | analysts; –                                               |
| Palma et al.<br>[32,34]       | incident-<br>management<br>event log          | alignment,<br>deviation cost<br>model          | n/d                       | auditor or<br>assessor; –                                 |
| DomainPrio<br>[12]            | one SOC’s in-<br>vestigations                 | population<br>analysis,<br>regression          | –                         | analysts; –                                               |
| SAT-SA (this<br>work)         | six categories<br>of submitted<br>SOC records | rule-based<br>workers,<br>saturating<br>fusion | peer cohorts,<br>ranking  | supervisor<br>decides;<br>signed<br>receipt and<br>ledger |

*Process conformance and peer comparison.* Conformance checking relates recorded behaviour to a model and exposes skipped or out-of-order activities [38,11,1], and has been applied to security audit trails [2] and to IT incident management against ITIL [15]. SAT-SA’s execution-gap and negative-space rules are rule-encoded conformance checks, not a new technique. Peer group analysis flags entities that diverge from similar entities [48,9]; SAT-SA uses median and median absolute deviation (MAD) [26] over a declared cohort.

*Supervision and human oversight.* Financial supervisors use technology to analyse submitted data (“suptech”) while keeping human judgement central [10]. Automation bias is a documented risk of decision support [36] and has been reviewed for SOC’s specifically [46]; we did not study it.

*Integrity of records.* Tamper-evident logs detect alteration of recorded history [40,13]; practical systems protect audit logs with trusted execution [31] or ratcheting [19], and post-quantum primitives have been applied to forensic logging [18]. SAT-SA proposes no logging scheme or cryptography; it composes SHA3-256 [28] and ML-DSA-65 [29] to bind one canonical supervisory record to the authorized decision.

*Gap.* Record-based incident assessment, supervisory analysis of submitted data, peer comparison and tamper-evident records each exist. We did not find a work that combines them for supervision of many SOC’s from their evidence submissions—with record-cited findings, cross-organization ranking and a cryptographically bound human decision—and evaluates the combination against simple baselines and on independent data. Given the search limits stated above, this is a low-confidence gap about a combination of established parts.

### 3 Problem Formulation

We formalize the workflow as it is implemented; no step below is learned from data.

*Submissions.* For a monitored organization (entity)  $e$  and assessment period  $t$ , a *submission*  $S_{e,t}$  is a versioned set of records partitioned into six evidence categories (alerts, cases, investigation steps, escalations, dispositions, assets). Each record  $r$  carries its source provenance (artifact digest

and record locator). Validation is a total function  $V(S_{e,t}) \in \{accept, reject\}$  that parses every record, resolves references (e.g. a step must reference an existing case) and enforces chronology; it is all-or-nothing, so a single invalid record rejects the version.

*Findings.* Each analytical worker  $w_j$ ,  $j = 1, \dots, 16$ , maps an accepted submission (and, for peer analysis, a declared cohort) to findings

$$f = (\rho_f, \text{subject}_f, E_f, s_f, \theta_f, c_f), \quad (1)$$

with rule  $\rho_f$ , subject, evidence references  $E_f \subseteq S_{e,t}$ , statistic  $s_f$ , threshold  $\theta_f$  and confidence  $c_f \in [0, 1]$ . A finding is a *signal* only if  $E_f \neq \emptyset$ ; findings without evidence references are withheld. Let  $F_e$  be the signal findings of the entity's run.

*Confidence.* Every signal carries a confidence vector with an analytical support  $a_f$ , an evidence completeness  $e_f$  and, for peer-based findings only, a peer confidence  $p_f$ ; the scalar used downstream is the weakest component,

$$c_f = \min(a_f, e_f, p_f). \quad (2)$$

Each worker sets these components by a fixed, hand-written rule (Tables 3 and 4): a constant, or a clipped linear function of the finding's effect size (e.g. how far a median closure time falls below its limit) and of record counts (e.g. the share of qualifying alerts, or the number of peers). The rules are heuristic: they were not fitted to data and are not calibrated probabilities, and sample size enters only through the count-based terms. When the evidence category a rule depends on was not submitted, its  $e_f$  is 0, so the finding contributes nothing to risk.

*Rule families.* Most execution-gap rules are *existence rules*: they fire when at least one qualifying record exists, e.g. a critical alert closed faster than a severity-specific limit. Peer deviation compares an entity metric  $x_e$  with the median  $\tilde{m}$  and median absolute deviation MAD of a declared cohort  $C$  (after symmetric 10% trimming when  $|C| \geq 4$ ) and fires when

$$|C| \geq 3 \quad \text{and} \quad |x_e - \tilde{m}| \geq 2 \text{MAD}(C); \quad (3)$$

within-entity anomalies use the same statistic with factor 3 over the entity's own values. MAD is unscaled.

*Risk fusion.* A fixed map  $\delta$  assigns each rule to one of 7 dimensions  $d$  with weights  $w_d$  (execution gap 25, peer deviation 20, detection gap 15, negative space 15, anomaly 15, investigation quality 5, escalation discipline 5; sum 100). With  $C_d = \sum_{f \in F_e, \delta(\rho_f)=d} C_f$ ,

$$R_d = w_d \frac{C_d}{1 + C_d}, \quad R_e = \min\left(100, \sum_d R_d\right). \quad (4)$$

Each dimension saturates at its weight. Two properties matter later:  $R_e$  is a heuristic score, not a probability or a severity; and it depends on summed confidences, not on how *prevalent* a problem is within the submission. Findings from different families naming the same subject are reported as corroborating clusters that do not change  $R_e$ .

*Prioritization.* Entities are ordered by the priority score

$$P_e = R_e + 10 \kappa_e + 5 \tau_e + 2 h_e, \quad (5)$$

where  $\kappa_e \in \{0, \frac{1}{4}, \frac{3}{5}, 1\}$  encodes the confidence bucket of the entity's findings,  $\tau_e \in [0, 1]$  decays linearly with run age over 30 days, and  $h_e$  counts high-severity signals. The coefficients are fixed defaults, not fitted.

*Decision and binding.* A supervisor (or organization administrator) records a decision  $D_e$  on the reviewed state. Finalization builds a canonical document  $doc_e$  (decision, findings, observations, risk, recommendations, submission records and their provenance), computes  $h_e = \text{SHA3-256}(doc_e)$ , signs  $h_e$  with ML-DSA-65, stores a receipt, and appends a ledger entry  $\ell_i$  whose hash is SHA3-256 of its canonical body including the predecessor hash  $\ell_{i-1}$ . Verification rebuilds  $doc_e$  from the live records and checks the digest, the decision binding, the signature and the chain.

*Questions.* Given this pipeline, we ask whether the declared mechanisms behave as specified under controlled conditions, how they compare with simple baselines, how they respond to imperfect evidence and cohort choice, what orchestration and integrity cost, and whether  $R_e$  tracks an independent outcome on real data (Section 5).

## 4 System Architecture and Analytical Methodology

Figure 1 shows the workflow as implemented. An analyst of the monitored organization uploads one file per evidence category through an HTTP

API; validation stores per-record provenance; a separate worker process claims queued runs under a lease and executes the analysis; a supervisor reviews and decides; and the decision is finalized and bound.

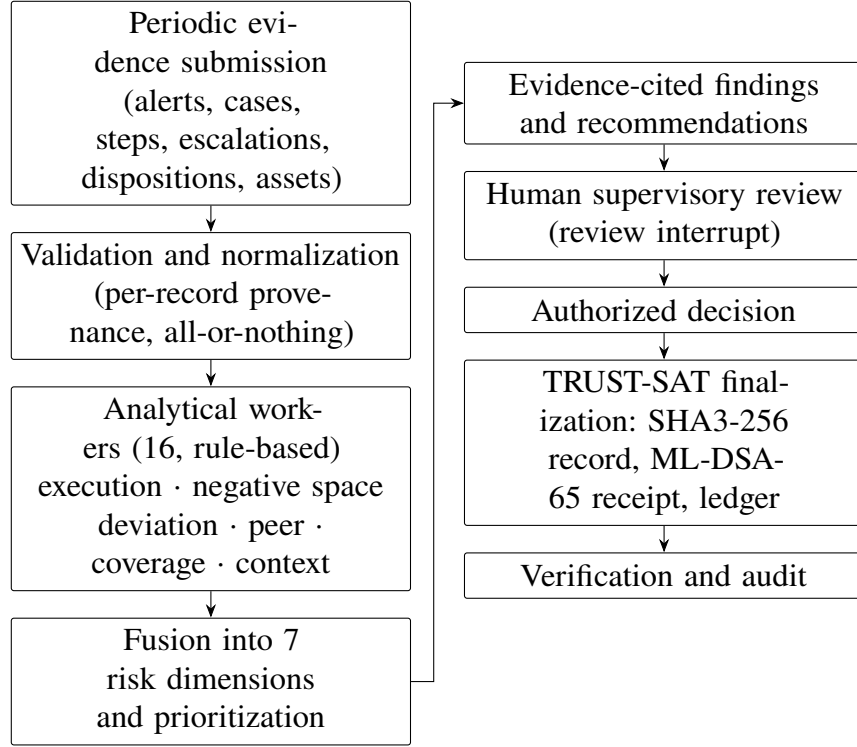


Fig. 1. SAT-SA supervisory workflow along the evaluated path. The analytical-worker stage is the evaluated subset of the registered agents (Table 2). All analytics are deterministic and rule-based; orchestration (direct worker or LangGraph) sequences the stages and pauses for the human decision.

#### 4.1 Architecture layers

The implementation registers 32 components that it calls *agents* (Table 2): 23 supervisory agents and 9 MLOps agents retained from the platform SAT-SA was built on. An agent here is a registered specification (purpose, inputs, outputs, evidence types) that maps to deterministic code; none is a

language-model or autonomous agent. Eleven supervisory agents cover the analysis families and group the 16 default analytical workers evaluated in this paper; the other twelve cover the pipeline around them, from ingestion and normalization through correlation, risk scoring, prioritization and recommendation to review, trust and provenance, reporting and validation. The reported experiments exercise the analytical workers and the pipeline from ingestion to trust; evidence assembly, meta-audit, report generation, the calibration-oriented validation agent and the retained MLOps agents are not evaluated. The count of 16 therefore refers to the analytical workers only, not to the whole architecture.

## 4.2 Analytical workers

The 16 default analytical workers fall into five functional groups. *Execution and process rules* (fast closure, acknowledgement without investigation, critical alert without escalation, repeated shallow investigation, recurrence without remediation, potential metric gaming) check individual records and short record chains against expected handling. *Negative-space analysis* detects the absence of expected work: cases without investigation steps, missing escalations or dispositions, critical assets without alerts; these are rule-encoded conformance checks. *Deviation analysis* comprises within-entity anomalies, drift and the peer rule of Eq. (3). *Coverage and completeness* (coverage gap, evidence completeness) flag unmonitored assets and missing evidence categories, and entity–asset resolution links records to the assets they concern. *Context workers* (workflow reconstruction, case similarity, cross-entity insights) reconstruct sequences and clusters; their signals also feed risk dimensions. Tables 3 and 4 list all 16 workers with the dimension each feeds and its confidence rule. Because a rule’s dimension is assigned by name prefix, the escalation-discipline and investigation-quality dimensions are fed only by workflow reconstruction (and by keyword-matching drift or cross-entity rules), while critical alerts without escalation count towards the execution gap. No worker uses a language model or a learned model.

## 4.3 Fusion, prioritization and recommendations

Signal findings are fused by Eq. (4) into an entity risk decomposed by dimension, so that a supervisor sees which findings produce which part of

Table 2. SAT-SA architecture inventory as implemented. “Agent” is the implementation’s name for a registered component; each maps to deterministic code, and none is a language-model or autonomous agent. Only the rows marked “yes” are exercised by the reported experiments.

| Layer                              | Count | Members                                                                                                                                                                              | Evaluated                              |
|------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Registered agents (agent registry) | 32    | 23 supervisory agents and 9 retained MLOps agents                                                                                                                                    | in part                                |
| Supervisory analytical agents      | 11    | one agent per analysis family; they group the 16 default analytical workers (the execution-gap agent groups six detectors)                                                           | yes                                    |
| Supervisory pipeline agents        | 12    | ingestion, normalization, correlation, risk scoring, prioritization, recommendation, review workflow, trust–provenance, evidence assembly, meta-audit, report generation, validation | ingestion to trust: yes; last four: no |
| Retained MLOps agents              | 9     | data, performance, security, quantum security, red team, training optimization, incident response, governance, optimization (inherited platform)                                     | no                                     |
| Default analytical workers         | 16    | listed in Tables 3 and 4                                                                                                                                                             | yes                                    |
| Orchestration graph nodes          | 5     | readiness, analysis, recommendations, human review, trust boundary (LangGraph mode)                                                                                                  | yes                                    |

Table 3. Default analytical workers evaluated in the reported experiments (part 1 of 2; the analytical subset of the registered agents, Table 2): risk dimension fed and confidence components,  $c_f = \min(a_f, e_f[, p_f])$  (Eq. (2)).  $q$  = qualifying records;  $N$  = records of the relevant category;  $\delta$  = effect size clipped to  $[0, 1]$ ;  $z$  = deviation in MADs. All constants are fixed in code. Dimensions: EG execution gap, NS negative space, AN anomaly, PD peer deviation, DG detection gap, IQ investigation quality, ED escalation discipline; kw = assigned by rule-name keyword (default AN).

| Worker                         | Dim. | Analytical support $a_f$                                                 | Evidence completeness $e_f$ (peer $p_f$ ) |
|--------------------------------|------|--------------------------------------------------------------------------|-------------------------------------------|
| Fast closure                   | EG   | $0.4 + 0.6 \delta$ ,<br>$\delta = 1 - \text{median/limit}$               | $q/N$ (alerts of that severity)           |
| Ack. without investigation     | EG   | $0.5 + 0.4 q/N$                                                          | $q/N$ (alerts)                            |
| Critical without escalation    | EG   | 0.8                                                                      | $q/N$ (alerts)                            |
| Repeated investigation         | EG   | $0.4 + 0.5 \delta$                                                       | $\min(1, N_{\text{steps}}/20)$            |
| Recurrence without remediation | EG   | 0.7                                                                      | $q/N$ (cases)                             |
| Potential metric gaming        | EG   | $0.4 + 0.4 \delta$                                                       | $\min(1, N/20)$                           |
| Negative space                 | NS   | 0.7 steps; 0.6 escalation, monitoring; 0.5 disposition; 0.4 low activity | 1 if category submitted, else 0           |
| Anomaly                        | AN   | $0.4 + 0.5 \min(1, z/3)$                                                 | 1                                         |
| Peer benchmark                 | PD   | $0.4 + 0.1 \min(1,  z /3)$                                               | 1; $p_f = \min(1, n_{\text{peers}}/10)$   |
| Drift                          | kw   | $0.4 + 0.6 \min(1,  \Delta_{\text{rel}} )$                               | 0.9/0.6/0.3 ( $\geq 3/\geq 1/0$ refs)     |

Table 4. Default analytical workers evaluated in the reported experiments (part 2 of 2); abbreviations as in Table 3.

| Worker                  | Dim.       | Analytical support $a_f$                              | Evidence completeness $e_f$ (peer $p_f$ )                   |
|-------------------------|------------|-------------------------------------------------------|-------------------------------------------------------------|
| Coverage gap            | DG         | $0.4 + 0.6 g$ ( $g$ unmonitored critical-asset share) | 0.9 if alerts link assets, else 0.4                         |
| Cross-entity insights   | kw         | $0.4 + 0.6 \min(1, \pi)$ ( $\pi$ prevalence)          | 0.9 if $\geq 3$ entities, else 0.6;<br>$p_f = \min(1, \pi)$ |
| Case similarity         | AN         | $0.5 + 0.5 s$ ( $s$ cluster share)                    | 0.7                                                         |
| Evidence completeness   | AN         | $0.4 + 0.6 m$ ( $m$ missing share)                    | $1 - m$                                                     |
| Workflow reconstruction | ED, IQ, AN | 0.85 / 0.8 / 0.6 by rule                              | 1                                                           |
| Entity-asset resolution | AN         | 0.6                                                   | 1                                                           |

the score. Entities are ordered by Eq. (5), and each finding family maps to a recommended supervisory action.

#### 4.4 Human supervisory review

Runs can require review. The system never decides: only a supervisor, or an organization administrator, can record the decision, after which the worker resumes. Automation assists; the supervisor retains terminal authority.

#### 4.5 Stateful orchestration

The same stages run in two modes. In *direct* mode the worker pauses the run after analysis and recommendations. In *graph* mode a LangGraph [25] graph with 5 nodes (readiness, analysis, recommendations, human review, trust boundary) checkpoints its state and interrupts at the review node. LangGraph provides checkpointing, interrupt/resume and recovery; it does not replace or alter the deterministic workers, and both modes produce identical findings and risk (Section 6).

#### 4.6 Integrity layer

After the decision, finalization binds the reviewed state as defined in Section 3: canonical serialization, a SHA3-256 commitment [28], an ML-DSA-65 [29] signature, a stored receipt and a hash-chained ledger entry in the spirit of tamper-evident logging [40,13]. The threat model is after-the-fact alteration of the supervisory record by someone with write access to the database or ledger file. Out of scope are compromise of the signing key, of the running process before finalization or of the operating system, and false records submitted by the monitored organization: TRUST-SAT binds what was analysed and decided, not whether the submitted records are true. An attacker who can rewrite the database and re-sign with the service key, or substitute a key pair consistently, is not detected unless the public key is pinned out of band.

### 5 Experimental Methodology

*Research questions.* RQ1: does SAT-SA emit the finding families declared for controlled scenarios? RQ2: how does it compare with simple baselines,

and what does each worker contribute? RQ4: how does it behave on imperfect evidence? RQ5: does verification detect alteration of the finalized record? RQ6: how sensitive is the peer rule to cohort size and dispersion? RQ7: what do durable orchestration and review cost, and do interrupted runs recover? EXT: does the pipeline run on independent real data, and is its risk associated with an outcome available there? RQ3, the utility of SAT-SA to supervisors, requires a human study that was not run. Table 5 maps every question to its experiment.

*Data.* All controlled data are *synthetic*. The *controlled catalog* contains 5 executable scenarios (of 10 catalog entries) authored with declared finding families and supervisory actions. Its labels are *construction labels*: they encode what a scenario was built to contain, so agreement shows that the mechanism responds to its designed inputs, not that it detects real misconduct. *Generated populations* (PR01) are 20 independently seeded populations of 20 entities, 5 of them declared pathological by the generator. *Engineered peer cohorts* (P01) are 96 cells crossing cohort size, dispersion and subject value. Controlled data were necessary because no public dataset contains SOC workflow records with supervisory ground truth. The single *external* dataset is the public UCI incident-management event log (dataset 498, Creative Commons Attribution licence) of one organization’s IT service desk [6,7]; it is IT incident data, not SOC data. Two security datasets were assessed and rejected: GUIDE has triage-graded SOC incidents but no acknowledgement, closure, investigation-step or escalation records [17], and CIC-IDS2017 contains network flows, not workflow records [42].

*Baselines.* Baselines are deliberately simple, because a supervisory tool must beat transparent rules to justify its complexity. For closure-time detection: fixed threshold, MAD rule, z-score and interquartile-range (IQR) rules, and seeded random. For prioritization: random order, critical-and-high alert volume, and a closure-speed heuristic that ranks entities by *fastest median closure time*. Externally: random, incident volume, reassignment rate and slowest median resolution; the last is close to the outcome’s own construct and serves as an upper reference, not a competitor.

*Metrics and statistics.* Precision, recall and F1 against construction labels; recall and normalized discounted cumulative gain (NDCG) [23] within

Table 5. Research questions, experiments and independent units. “Trials” repeat a measurement on the same unit and are not independent.

| RQ             | Exp.      | Data (ground truth)                   | Baselines                          | Independent unit (n)         |
|----------------|-----------|---------------------------------------|------------------------------------|------------------------------|
| RQ1 detection  | C01       | catalog (construction labels)         | –                                  | scenario (5)                 |
| RQ2 baselines  | C01       | closure corpus (construction)         | threshold, MAD, z, IQR, random     | record (12)                  |
| RQ2 ranking    | PR01      | generated populations (generator)     | random, volume, fastest closure    | population (20)              |
| RQ2 ablation   | A01       | catalog; populations                  | full worker set                    | scenario (5); population (5) |
| RQ4 robustness | R01b      | perturbed catalog (declared contract) | –                                  | fixture (5); 245 conditions  |
| RQ5 integrity  | T01       | one workflow (declared outcome)       | unsigned-field control             | workflow (1); 14 mutations   |
| RQ6 peer       | P01       | engineered cohorts (design)           | –                                  | designed cell (96)           |
| RQ7 cost       | O01       | catalog scenario                      | direct vs. graph vs. unreviewed    | 1 machine; 30 paired trials  |
| RQ7 recovery   | O02       | catalog scenario (reference run)      | uninterrupted run                  | 12 cells; 36 runs            |
| EXT            | X02b, X03 | UCI-498 (SLA flag)                    | random, volume, reassign., slowest | group of 1 organization (50) |

the top  $k \in \{10, 20, 30\}$ % of entities; review volume needed to find all pathological entities; conformance with the declared validation contract; wall-clock seconds and database statements; and Spearman  $\rho$  against the share of incidents that missed their service-level agreement (SLA). We report medians, 95th percentiles only when  $n \geq 20$ , and seeded percentile bootstrap intervals [14] only when there are at least ten independent units; comparisons are paired by seed or trial with counts of units where one arm was higher, equal or lower. No  $p$ -values are computed and no significance is claimed; all intervals are descriptive and uncorrected. Table 5 distinguishes repeated trials from independent units.

*Evidence discipline.* Every experiment writes a write-once bundle (manifest with code commit and a dirty-tree flag, configuration, raw results, processed metrics) with SHA-256 hashes. The canonical bundles are frozen and hash-verified; superseded bundles (for example X02, replaced by X02b after a bookkeeping correction that left all analytical outputs identical) are kept but not cited. Every number in this paper is rendered from the frozen bundles; no detector, threshold or weight was changed on evaluation data.

## 6 Results on Controlled Data

### 6.1 Detection and closure-time baselines (RQ1, RQ2)

Over the 5 catalog scenarios analysed in one shared database, SAT-SA emitted every declared finding family (true positives 5, false negatives 0) and 7 undeclared families: micro precision 0.417, recall 1.000, F1 0.588. The recommended supervisory action matched the declared action in 4 of 5 scenarios. Precision depends on database layout: analysed one scenario per database (A01), the full worker set reached 0.556, because cross-entity workers in a shared database see other scenarios’ records.

On 12 construction-labelled critical closures (3 fast), SAT-SA’s fast-closure detector, a fixed threshold and a MAD rule each reached F1 1.00; z-score and IQR rules found none (recall 0.0); seeded random reached 0.33. *The detector ties simple baselines and offers no detection advantage on this corpus.*

## 6.2 Prioritization against simple baselines (RQ2)

Across 20 generated populations, SAT-SA placed a median 0.800 of the pathological entities in its top 20% and needed a median of 7.0 reviews to find all of them (Table 6, Fig. 2). Its recall@20% exceeded random order in all 20 populations (median paired difference 0.59, 95% bootstrap 0.41 to 0.60) and alert-volume ranking in all 20 (0.40; 0.30 to 0.60). *Against the fastest-median-closure heuristic the comparison is mixed*: higher in 8, equal in 11 and lower in 1 populations (median difference 0.00, interval 0.00 to 0.20); SAT-SA needed fewer reviews in 12 populations and more in 6.

These results must be read against how the data were made. The populations come from SAT-SA’s own generator, whose pathological profile deliberately injects the behaviours that SAT-SA’s detector families target, so the construction favours SAT-SA. Because the generated populations inject behaviours targeted by SAT-SA’s detector families, PR01 is a controlled sensitivity and ranking-mechanism evaluation rather than an unbiased benchmark of general supervisory effectiveness. It shows that the ranking responds as designed under these generated conditions; it does not establish ranking superiority in general or effectiveness in real SOC’s. Even under this favourable construction, a single closure-speed rule captures most of the separation.

Table 6. Prioritization on 20 generated populations of 20 entities (5 pathological): medians over populations (PR01). P@10 = precision in the top 10%; R@20 = recall in the top 20%; N@20, N@30 = NDCG in the top 20% and 30%; Vol. = reviews needed to find all pathological entities. The generator injects behaviours SAT-SA targets, which favours SAT-SA by construction.

| Ranking                       | P@10  | R@20  | N@20  | N@30  | Vol. |
|-------------------------------|-------|-------|-------|-------|------|
| SAT-SA ( $P_e$ )              | 1.000 | 0.800 | 1.000 | 0.869 | 7.0  |
| Fastest median closure        | 1.000 | 0.600 | 0.832 | 0.837 | 11.5 |
| Critical/high alert volume    | 0.500 | 0.200 | 0.246 | 0.318 | 18.0 |
| Random (mean of permutations) | 0.245 | 0.201 | 0.248 | 0.276 | 17.5 |

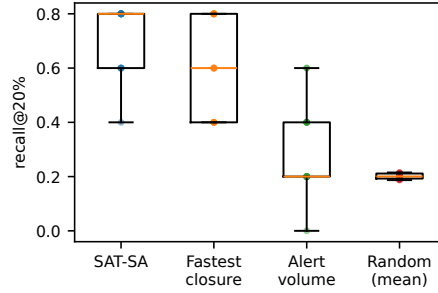


Fig. 2. Recall@20% of generator-labelled pathological entities across 20 generated populations (PR01). The generator injects behaviours that SAT-SA targets, so this is a controlled mechanism test, not an unbiased benchmark.

### 6.3 One-worker ablation (RQ2)

Removing one of the 16 workers at a time (Table 7), fast-closure and negative-space carried recall on the catalog (0.600 and 0.600 without them; 1.000 with all workers). Removing recurrence-without-remediation or coverage-gap *raised* precision because those workers emitted only undeclared families here. On five generated populations (exploratory), mean recall@20% was 0.760 with all workers and rose to 0.800 without the anomaly, recurrence, coverage-gap or workflow-reconstruction worker, whereas removing the peer-benchmark or case-similarity worker lowered it to 0.720. On these populations, therefore, some workers add ranking noise to the measured metric.

We read this nuanced result narrowly. The full worker set spans a broader analytical signal space than the generated populations exercise, and the ranking objective is specific to the generator’s pathological profile. The ablation therefore shows that some workers can reduce this particular metric; it does not show that those workers are unnecessary, that they would be harmful in real SOC environments, or that all 16 workers are necessary. Worker utility is task- and population-dependent, which is one more reason why external and human evaluation are needed.

Table 7. One-worker ablation (A01): catalog micro precision/recall (5 scenarios, one database per scenario) and mean population recall@20% (Pop. R@20; five seeds, exploratory). Only workers whose removal changed a value are listed; “–” = unchanged from the full set.

| Workers                             | Precision | Recall | Pop. R@20 |
|-------------------------------------|-----------|--------|-----------|
| All 16                              | 0.556     | 1.000  | 0.760     |
| – fast closure                      | 0.429     | 0.600  | 0.600     |
| – negative space                    | 0.500     | 0.600  | 0.640     |
| – evidence completeness             | 0.500     | 0.800  | –         |
| – recurrence without<br>remediation | 0.714     | 1.000  | 0.800     |
| – coverage gap                      | 0.625     | 1.000  | 0.800     |
| – anomaly                           | –         | –      | 0.800     |
| – workflow reconstruction           | –         | –      | 0.800     |
| – peer benchmark                    | –         | –      | 0.720     |
| – case similarity                   | –         | –      | 0.720     |

#### 6.4 Imperfect evidence (RQ4)

Each perturbed condition declared its family, affected records and expected validation outcome before it ran, on a fresh tenant database. Validation matched the declared contract in 245 of 245 conditions with an expectation (Table 8): malformed timestamps, chronology violations, missing columns and duplicates with native identifiers were rejected. Two behaviours limit interpretation (Fig. 3). First, validation is all-or-nothing: independent omission of records left dangling references and rejected the whole submission in 3, 11 and 14 of 25 replicates at 10, 25 and 50% omission, whereas referentially consistent omission was never rejected and changed the emitted finding families in 1, 11 and 15 replicates. Second, *stale and contradictory evidence is accepted silently*: all 15 conditions with records outside the assessment period were accepted without warning and without any change to findings or risk, and contradictory dispositions or a closed case with an open alert produced no finding. SAT-SA implements no period, recency or cross-record consistency validation.

Table 8. Imperfect-evidence conditions by family (R01b). Rejected = whole submission rejected by validation; changed = emitted finding families differed from the paired control; “–” = no condition rejected (all analysed); \*no declared expectation. Validation matched its declared contract in 245 of 245 conditions with an expectation.

| Family       | $n$ | Rejected | Analysed | Changed |
|--------------|-----|----------|----------|---------|
| Missingness  | 173 | 33       | 140      | 43      |
| Duplication  | 34  | 25       | 9        | 5       |
| Malformation | 15  | 15       | 0        | 0       |
| Conflict     | 18  | 5        | 13       | 4       |
| Staleness*   | 15  | –        | 15       | 0       |
| Control      | 5   | –        | 5        | 0       |

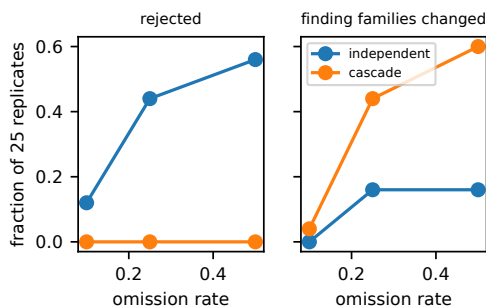


Fig. 3. Record omission (R01b): independent omission is often rejected by all-or-nothing validation; referentially consistent omission is accepted and changes findings more often as the rate grows.

### 6.5 Peer-cohort sensitivity (RQ6)

Over 96 engineered cells (Fig. 4), cohorts of two peers never produced a finding, as Eq. (3) requires three. With a tight cohort spread, 7 of eight subject values were flagged from three peers upward—every value except the one at the cohort median. With a wide spread only extreme slow subjects were flagged at small cohorts (2 at three peers), and fast subjects only from five peers upward as the MAD narrowed (4 at five peers). The peer rule is therefore sensitive to cohort size and dispersion: small or heterogeneous cohorts miss deviations, and tight cohorts flag small ones.

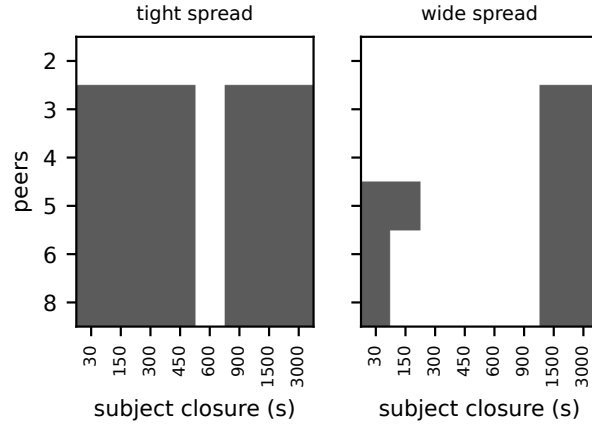


Fig. 4. Peer findings by cohort size, spread and subject closure value; dark cells emitted a finding (P01, 96 cells).

### 6.6 Orchestration cost and recovery (RQ7)

On one machine with SQLite, 30 paired trials per mode gave median processing times, excluding the human decision, of 1.276 s (direct), 1.433 s (LangGraph) and 0.976 s (unreviewed benchmark mode) (Table 9). LangGraph minus direct was a median 0.096 s per workflow (95% bootstrap −0.020 to 0.211 s; the interval includes zero), with 38 more database statements and 7 checkpoints per run; findings and risk were identical across modes in every trial. Review with finalization added a median 0.318 s over unreviewed processing; finalization took 0.288 s and receipt verification

0.062 s. Interruptions were injected at 6 points in both modes with 3 trials per cell: transient failures and simulated crashes during analysis, before recommendations, at a worker restart during review, and during finalization. *In the evaluated interruption scenarios*, 36 of 36 runs recovered, 36 with outputs identical to the uninterrupted reference; completed stages were repeated 0 times, and every run had exactly one decision, one finalization and a verifying receipt. Real process kills, network partitions and database failover were not exercised.

Table 9. Orchestration cost on one machine with SQLite (O01, 30 paired trials per mode; the human decision excluded). Median and p95 in seconds; Outside = seconds outside the domain stages; DB = database statements. Graph – direct: median 0.096 s, 95% bootstrap –0.020 to 0.211 s (graph slower in 19 of 30 pairs). Finalization took 0.288 s, receipt verification 0.062 s and run and finding attestation 0.853 s (medians).

| Mode                      | Median | p95   | Outside | DB  |
|---------------------------|--------|-------|---------|-----|
| Direct (reviewed)         | 1.276  | 2.010 | 0.055   | 420 |
| LangGraph (reviewed)      | 1.433  | 2.206 | 0.115   | 458 |
| Unreviewed benchmark mode | 0.976  | 1.655 | 0.029   | 253 |

## 6.7 Integrity of the decided record (RQ5)

On one finalized controlled workflow, each of 14 mutations was applied, verified, restored and re-verified (Table 10). Verification detected 13 of 13 mutations of signed state—decision action and reason, finding, risk, recommendation, observation scope, source provenance, submission record payload, artifact digest, canonical payload, receipt signature, receipt public key and ledger chain—while the negative control (1 mutation of an unsigned operational queue field) still verified, as designed. These are detections of the tested mutation classes on one workflow, not a detection rate, and detection is not prevention (Section 4).

Table 10. Integrity mutation matrix on one finalized workflow (T01): 13 of 13 mutations of signed state detected; the unsigned control still verified. Full targets are in the supplementary material.

| Mutated element                                 | Expected | Observed |
|-------------------------------------------------|----------|----------|
| Decision action; decision reason                | tampered | tampered |
| Finding rationale; observation scope            | tampered | tampered |
| Risk profile; recommendation                    | tampered | tampered |
| Source provenance; submission record<br>payload | tampered | tampered |
| Artifact digest; canonical payload              | tampered | tampered |
| Receipt signature; receipt public key           | tampered | tampered |
| Ledger chain (predecessor hash)                 | tampered | tampered |
| Unsigned queue field (control)                  | verified | verified |

## 7 External Evaluation and Failure Analysis

*Setup.* The unchanged pipeline analysed the UCI-498 incident log [6], a real ServiceNow event log of one organization’s IT service desk, with each final assignment group of sufficient size treated as an entity: 50 groups, 22,604 of 24,918 incidents, 47,420 investigation-step proxies derived from work-state events and 19,656 escalation proxies derived from reassignments. The only outcome available is the final SLA flag. SLA labels were loaded only after SAT-SA’s ranking existed and never entered a submission. *This is IT incident data, not SOC data*; it is, however, the log on which prior work demonstrated incident-management compliance assessment [33] and modelled SLA attainment [44].

*Feasibility.* All 134,888 submitted rows were accepted (0 rejected) and every group’s analysis completed, with a median of 9 s per group.

*Negative result.* SAT-SA’s risk  $R_e$  was not associated with the SLA-miss rate: Spearman  $\rho = -0.11$ , 95% bootstrap  $-0.42$  to  $0.21$  (Table 11, Fig. 5). Slowest median resolution reached  $\rho = 0.94$  ( $0.86$  to  $0.97$ ), as expected for a timeliness outcome; reassignment rate reached  $0.37$ . Among the 12 groups in the top SLA-miss quartile, SAT-SA’s precision@10% was  $0.40$  (random  $0.25$ , slowest resolution  $1.00$ ), and it needed to review 50 groups to find all of them (slowest resolution 24).

Table 11. UCI-498 IT incident log (X02b, X03): Spearman association of group scores with the SLA-miss rate over 50 assignment groups of one organization (95% seeded bootstrap intervals). Not SOC data.

| Group score                                 | Spearman $\rho$ | 95% interval   |
|---------------------------------------------|-----------------|----------------|
| SAT-SA risk $R_e$                           | -0.11           | -0.42 to 0.21  |
| Incident volume                             | -0.15           | -0.42 to 0.16  |
| Reassignment rate                           | 0.37            | 0.08 to 0.60   |
| Slowest median resolution (upper reference) | 0.94            | 0.86 to 0.97   |
| Execution-gap dimension of $R_e$ (X03)      | -0.41           | -0.66 to -0.11 |
| Fast-closure prevalence (X03)               | -0.70           | -0.82 to -0.52 |

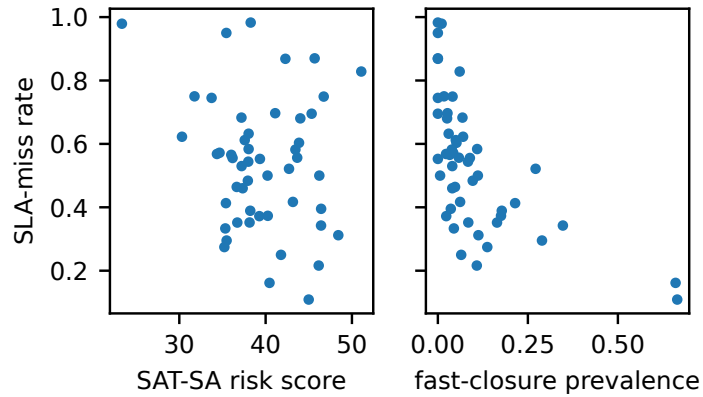


Fig. 5. UCI-498 IT incident log (not SOC data): SAT-SA risk (left) and fast-closure prevalence (right) against the SLA-miss rate; one point per assignment group (50 groups; X02b, X03).

*Failure analysis.* A diagnostic analysis (X03) on the same data changed no detector, threshold or weight. It attributes the result to four causes. (i) *Construct mismatch (primary).* On a service desk, fast resolution is the goal: the prevalence of fast closures was *negatively* associated with SLA misses ( $\rho = -0.70, -0.82$  to  $-0.52$ ). SAT-SA treats fast closure of a security alert as possible skipped investigation, so its execution-gap dimension, which this signal drives, ran opposite to the outcome ( $\rho = -0.41, -0.66$  to  $-0.11$ ). SLA timeliness and supervisory execution quality are different constructs, and here they move in opposite directions. (ii) *Detector saturation.* 4 detector families fired for every group although their input conditions varied widely—cases without any step 1.7% to 78.4% of cases, medium-or-higher alerts with fewer than two steps 5.2% to 85.9%. Existence rules cannot discriminate submissions with thousands of tickets, and Eq. (4) adds near-constant amounts for them because it ignores prevalence; risk still took 49 distinct values and was unrelated to volume ( $\rho = 0.02$ ). (iii) *Missing supervisory dimensions.* 3 of the 7 risk dimensions had no input data; investigation steps and escalations are proxies (state changes and reassignments), and monitoring coverage is unavailable. (iv) *A minor adapter artifact.* The log has no step notes; the adapter submits empty notes, which satisfy a repeated-investigation rule in 39 groups. The analysis found no temporal or label defect: only 3 of 7,907 SLA misses occurred while another group held the ticket, and reassigned incidents missed more often (53.8% versus 21.5%), consistent with the positive reassignment association.

*Decision.* No risk-model change was made. Changing detectors or weights to raise the correlation with SLA misses would optimize SAT-SA toward a timeliness construct it does not claim to measure, using the evaluation data for tuning. We therefore read X02b and X03 as *feasibility and construct-validity evidence*, not as external validation of effectiveness: on data whose outcome measures a different construct and that lacks investigation, escalation and coverage semantics, SAT-SA’s supervisory signals did not transfer.

## 8 Discussion

*Multi-signal supervision versus simple heuristics.* The controlled results show the mechanisms working as specified, but they do not show that the

multi-worker design adds value over transparent rules: fixed-threshold and MAD rules tie the fast-closure detector, a fastest-median-closure ranking equals or nearly equals SAT-SA in most generated populations—populations that favour SAT-SA by construction—and four workers *lowered* population recall in the ablation. This is consistent with a generator whose pathological profile includes fast closures; more elaborate workers can only earn their place on data where pathologies are not separable by one rate, and the evidence here neither establishes nor rules out that they do. For a supervisor, a small set of transparent rates may be a reasonable first instrument, with record-cited findings serving as the explanation layer rather than the ranking signal.

*Mechanism validation versus external validity.* Controlled experiments with construction labels answer whether each mechanism does what it declares; they cannot answer whether its output reflects supervisory quality. The external evaluation shows why the distinction matters: the same fast-closure signal that is a warning in a SOC is a success indicator in a service desk, so a construct defined for one setting inverts in the other. The methodological lesson generalizes beyond SAT-SA: an external outcome used to “validate” a supervisory score must measure the same construct, and a negative association with a different construct is evidence about constructs, not a defect to tune away. The external study also exposed a design limitation within SAT-SA’s own domain—existence rules fused by summed confidence saturate on large submissions—which calls for prevalence-aware scoring developed on held-out SOC data.

*Orchestration versus overhead.* On this workload, checkpointed orchestration cost a median 0.096 s per workflow with an interval that includes zero, at the price of 38 additional database statements; in exchange, all evaluated interruptions recovered without duplicated decisions. Whether this trade holds under load, on PostgreSQL or across machines was not measured.

*Integrity versus cost.* Finalization and verification are sub-second on this machine, and the tested alterations of the decided record were detected. The guarantee is narrow by construction: it binds what was analysed and decided, and it is only as strong as key custody, which was not evaluated.

*Peer analysis versus cohort dependence.* Peer comparison is the most supervisor-specific signal, but P01 shows that its output is a function of cohort size and spread as much as of the subject. Cohort definition is therefore a supervisory decision that should be recorded and reviewed with the finding.

*The human in the loop.* SAT-SA is decision support. Automation bias is a documented risk for such tools [36,46]; without a study with supervisors we cannot say whether record-cited findings help or anchor their judgement.

## 9 Threats to Validity and Limitations

*Construct validity.* Controlled labels are construction labels written by the system’s developers. The external outcome (SLA miss) measures timeliness, not supervisory execution quality, and investigation and escalation are proxies there. Risk scores have no validated meaning as probabilities or severities.

*Internal validity.* Populations for prioritization and ablation come from SAT-SA’s own generator, which injects the behaviours its detectors target and favours SAT-SA; the ranking results are therefore controlled mechanism evidence, not an unbiased benchmark. Precision depends on database layout. Thresholds and confidence rules were set by design, never fitted, and never tuned on independent development data.

*Worker complexity.* The one-worker ablation does not show that all 16 analytical workers are necessary or that the set is optimal: on the generated populations, removing some workers raised the ranking metric. Whether any worker helps or harms on real supervisory data is untested.

*External validity.* The evaluation is predominantly synthetic and controlled. There is no SOC data from a real organization; the external dataset is one IT service desk; generated populations have 20 entities, which makes top- $k$  estimates coarse; the ablation population analysis uses five seeds.

*Statistical conclusion validity.* Intervals are descriptive, seeded and uncorrected across many comparisons; no hypothesis is tested. Timing results are repeated runs on one machine and do not generalize across hardware, databases or load. Recovery was tested with injected exceptions at worker boundaries, not real process kills, partitions or database failover.

*Human factors.* A human-evaluation protocol exists but *no study was executed*: there are no participants and no usability or decision-quality results. An expert-label pipeline exists but *no expert-labelled data exist*.

*Systems and deployment validity.* All measurements used SQLite and local storage on one machine. The local API and worker processes passed 18 HTTP smoke checks (0 failed). Docker was not available, so the container image and compose topology were not executed; SeaweedFS object storage, the CI topology job and any hosted deployment were not executed. PostgreSQL played no part in any reported measurement. (After the evidence freeze, the schema migrations and the database test suite were run against a local PostgreSQL instance during maintenance; this is not part of the evaluation.) No claim of production readiness or scale is made.

## 10 Reproducibility and Artifact Availability

*Evidence chain.* Each reported number is generated from a canonical bundle in a hash-verified evidence freeze; the build verifies bundle hashes before reading them, and an audit regenerates all paper values, tables and figures and compares them byte for byte, rejects hand-typed decimals and checks every citation against a verified literature matrix.

*Reproduction levels.* We claim only what was demonstrated. *Level 1, artifact verification (demonstrated)*: every recorded hash is recomputed, in the working repository and in fresh clones; edited manifests, raw results, metrics and configurations are rejected. *Level 2, result regeneration (demonstrated)*: paper values, tables and figures regenerate byte-identically from the frozen bundles. *Level 3, experiment reproduction (demonstrated for two experiments)*: re-running the integrity mutation matrix (T01) and the peer sweep (P01) under new experiment identifiers reproduced their analytical outputs; timing experiments reproduce only in distribution. *Level 4, environment reproduction (not demonstrated)*: a container image and compose topology exist but were not executed; all runs used one Windows machine. *Level 5, external-data reproduction (demonstrated once, same machine)*: the external experiment was executed twice from the checksum-verified public dataset with identical analytical outputs; there is no third-party reproduction.

*Code and data availability.* The source code, experiment runners, evidence freeze, paper data layer and literature-search scripts are public under the MIT licence at <https://github.com/PrathamKapoor/SAT-SA-with-PQC>. The reported evaluation is identified by commits, not by a branch: each experiment was executed from the source commit recorded in its bundle manifest (listed in the supplementary material), and the frozen evidence set `satsa-evidence-freeze-v2` (canonical code commit `837d1ef`) was committed as `85b7763`. All controlled data are synthetic and regenerate from the recorded seeds and configurations. The external dataset [6] is public under a Creative Commons Attribution (CC BY) licence and is not redistributed; the experiment downloads it and verifies its checksum. No personal data, no data from a real SOC and no human-participant data were used.

## 11 Conclusion

We evaluated record-level supervisory analytics for SOC as implemented in SAT-SA: rule-based analysis of the records a monitored organization submits, saturating risk fusion and ranking, a human supervisory decision, and cryptographic binding of that decision to the analysed state. Under controlled conditions the mechanisms behave as specified—declared findings are emitted, validation follows its contract, the tested alterations of the decided record are detected, and the evaluated interruptions recover—but simple rules match much of the detection and ranking performance even on generated populations that favour SAT-SA by construction, not every analytical worker improved the ranking metric, and stale and contradictory evidence passes silently. On a public IT incident log the supervisory risk did not track the only outcome available, because that outcome measures timeliness, which the fast-closure signal treats in the opposite direction, and because existence rules saturate at scale. What can be concluded is therefore mechanism-level: the pipeline is inspectable and behaves as designed, while its usefulness for supervision is untested. The evidence needed next is not more controlled experiments but SOC records with independent outcomes held out from development, expert-labelled findings, a study with supervisors, prevalence-aware scoring and consistency checks for stale and conflicting records, and validation of the containerized PostgreSQL deployment.

**Ethics.** No human participants, personal data or data from a real security operations centre were used in this study.

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